

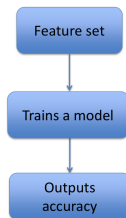
Overview

- Motivation for dimensionality reduction
- Feature selection
 - Wrappers
 - Filters
 - Embedded methods
- Feature transformation
 - Principal Component Analysis (PCA)
 - Autoencoders
 - Clustering

Feature selection

Wrappers

- Rely on a feature search strategy to find an “optimal” subset of features based on the performance of the classifier
- **Pros**
 - High accuracy
 - Specific to the classifier of interest
- **Cons**
 - Computationally expensive



Feature selection

Wrappers - Possible feature search strategies

- Exhaustive search
 - Try all possible feature combinations
 - M features $\rightarrow 2^M$ possible subsets
- Sequential forward selection
 - Greedy incremental selection of best performing features
- Recursive backward elimination
 - Starting from the full feature set, greedy selection of features which hurt performance
- Genetic algorithms
 - Random selection of features
 - Update of feature selection probabilities based on performance metrics

Feature selection

Wrappers: Sequential Feature Selection

- Cost is $M + (M - 1) + \dots + 1 = \frac{M(M+1)}{2}$, instead of 2^M

